

ECON1010

Introductory Microeconomics

Tutorial 11 | Public Goods



This Week in ECON1010

CML 5 is due **tomorrow** 4.00pm May 19

CML 6 opens **this week** 9.00am May 20

**Tutor
evaluationssssssssssssssssssssssssssssss!!!!**



Assessment

85% **7**

75% **6**

65% **5**

50% **4**

47% **3**

30% **2**

0% **1**

CMLs (25%)

- Computer Managed Learning
- Each worth 5%
- Best 5/6 count towards final grade

CLEAR-JE (20%)

- Contextualised Learning of Economics via Authentic Reflection using Journal Entries
- Each worth 10%
- Best 2/3 count towards final grade

Final Exam (55%)

Tools and Resources

I need HELP...

1. **Ed Discussion Board**
(Blackboard)
2. **Consultation** (8.00am
Mon, 39-125A)
3. **Staff email**
econ1010@uq.edu.au

I REALLY need HELP...

- <https://my.uq.edu.au/contact/student-central>

ECON1010 Consult Timetable

Semester 1 2026

Time	Mon	Tue	Wed	Thu	Fri
8:00-8:30	CONSULT - SAM 39-125A				
8:30-9:00					
9:00-9:30					
9:30-10:00					
10:00-10:30				CONSULT - LUCAS 39-125A	
10:30-11:00					
11:00-11:30	CONSULT - GABE 39-125A	CONSULT - BORIS 39-125A	CONSULT - IKMAL 39-125A	CONSULT - NIK 39-125A	CONSULT - CARL 39-641
11:30-12:00					
12:00-12:30					
12:30-13:00					
13:00-13:30					LECTURE
13:30-14:00		CONSULT - DYLAN 39-125A			
14:00-14:30				CONSULT - GUANCHU 39-125A	
14:30-15:00					
15:00-15:30					
15:30-16:00					
16:00-16:30			CONSULT - KIRTAN 39-125A	CONSULT - NANDAN 39-125A	
16:30-17:00					
17:00-17:30					

Student Central

Need help? We've got you covered.

Contact us

Monday to Friday

- ✉ Email
- ☎ 1300 275 870 (8.30am-5pm)
- 📍 Building 42, St Lucia
- 📍 Building 8101A, Gatton

Chat – 1:51 (1 Minute) wait

Emergency help

For immediate risk:

UQ Campus Security
☎ 07 3365 3333 (24/7)

Off-campus emergency
☎ 000 (24/7)

Crisis support

For urgent mental health support:

UQ Counselling and Crisis Line
☎ 1300 851 998 (24/7)

Text a Crisis Counsellor
☎ 0488 884 115 (4.30pm-8am)

Sexual Misconduct Support Unit (SMSU)

Public Goods

Question 1a

In your own words (and using a creative example that could be found at a university campus) what are:

- i) Rivalrous goods
- ii) Excludable goods
- iii) Public goods
- iv) Collective goods
- v) Common goods
- vi) Private goods
- vii) Tragedy of commons



Question 1a

i) Rivalrous goods

Goods that are worsened when used by many people.

Examples: The university libraries – inaccessible when full and less productive when noisy. UQ gym - too many people reduce the number of exercises / machines available.

ii) Excludable goods

Goods which are not free to the general public.

Examples: UQ gym - access to paying members only. Paid parking meters – must pay to park (otherwise fined or towed).



Question 1a

iii) Public goods

Goods available to the public that can be shared freely without problems (ie nonexcludable and non-rivalrous)

Examples: Watching the sunset across the UQ lakes – available to the public for free (non-excludable) and others' viewing it doesn't reduce your enjoyment (non-rivalrous). Sitting on the UQ Great Court – free to the public to access (non-excludable) and others' use will not affect your enjoyment (non-rivalrous, note: when there are +5000 people using The Great Court, it becomes rivalrous as the available space to sit reduces and the grass, becomes damaged).

iv) Collective goods

Goods only available to some but can be shared without problems (ie excludable and nonrivalrous).

Example: Watching a lecture online – limited to those enrolled in the class (excludable) and may be downloaded or watched without obstructing others' use (non-rivalrous).

Question 1a

v) Common goods

Goods that are available to everyone but are made worse when used too much (ie nonexcludable and rivalrous).

Examples: Public microwaves in the food courts – free to use by the public (non-excludable) and less desirable when used by many people due to longer wait times, dirty interior, and more likely to malfunction (rivalrous). UQ libraries – open to the public during the day (non-excludable) but less desirable when full as there are no seats available and increased noise pollution (rivalrous).

vi) Private goods

Goods that are not given away freely to everyone nor able to be freely shared (ie excludable and rivalrous).

Example: A Guzman and Gomez burrito bowl – costs money (excludable) and when shared there is less to eat yourself (rivalrous). Playing billiards (pool) in the Red Room uni bar – limited to paying players (excludable) and only available to 2 or 4 players per game (rivalrous).

Question 1a

vii) Tragedy of the commons

All Common Goods – as the goods' free use naturally encourages overuse (to the detriment of others' use).

Examples: Public microwaves UQ Libraries



Question 1b

Complete the table to indicate the definition of collective, common, public and private goods. Include the examples you identified in parts i) to iv) above.

	Excludable	Non-excludable
Rivalrous		
Non-rivalrous		

Question 1b

Complete the table to indicate the definition of collective, common, public and private goods. Include the examples you identified in parts i) to iv) above.

	Excludable	Non-excludable
Rivalrous	Private Burrito bowl Billiards game at bar	
Non-rivalrous		

Question 1b

Complete the table to indicate the definition of collective, common, public and private goods. Include the examples you identified in parts i) to iv) above.

	Excludable	Non-excludable
Rivalrous	Private Burrito bowl Billiards game at bar	
Non-rivalrous	Collective/Club Watching lectures online	

Question 1b

Complete the table to indicate the definition of collective, common, public and private goods. Include the examples you identified in parts i) to iv) above.

	Excludable	Non-excludable
Rivalrous	Private Burrito bowl Billiards game at bar	Common Public microwaves UQ library
Non-rivalrous	Collective/Club Watching lectures online	

Question 1b

Complete the table to indicate the definition of collective, common, public and private goods. Include the examples you identified in parts i) to iv) above.

	Excludable	Non-excludable
Rivalrous	Private Burrito bowl Billiards game at bar	Common Public microwaves UQ library
Non-rivalrous	Collective/Club Watching lectures online	Public UQ Lakes sunset Sitting in Great Court

Question 2

- a) Emily and George are the only two residents in a neighbourhood and they share the same driveway. They would like to have the driveway paved. Emily would be prepared to pay \$1500 on her own to pave the driveway. George would be happy to pay \$1000 on his own to pave the driveway. Irrespective of who pays for the paving, both people will benefit from it.
- i) What is the most a contractor can charge for paving the driveway and still be hired to complete the work?
 - ii) The cost of paving the driveway is \$2000. It is EFFICIENT / INEFFICIENT for them to share this cost in some way, because then they will have a total economic SHORTAGE / SURPLUS of \$ ____.
- b) Emily and George are now considering how to divide the payment using the 'tax' methods they learned about in microeconomics. They discuss a Head Tax (equal price per person) and a Proportional Tax (% of use or income). George earns half the salary of Emily but uses the driveway twice as much.
- i) Which tax will Emily want?
 - ii) Which tax will George want?
 - iii) Which taxes will result in the construction of the driveway if it costs \$2000?



Question 2a

i) What is the most a contractor can charge for paving the driveway and still be hired to complete the work?

The contractor can charge a maximum of \$2500. This is the most Emily and George will pay when combining their willingness to pay.

*ii) The cost of paving the driveway is \$2000. It is **EFFICIENT / INEFFICIENT** for them to share this cost in some way, because then they will have a total economic **SHORTAGE / SURPLUS** of \$ ____.*

Efficient; Surplus; \$500



Question 2b

i) Which tax will Emily want?

Emily will receive a greater net benefit if she pays less. Emily will prefer the Proportional Tax (% of use) because she only uses the driveway half as much as George. This means she will only pay for 33% and George pays 66%.

Note: it is possible that Emily will not know how much George will use the driveway and instead vote for a Head Tax (equal 50/50 split of costs).

ii) Which tax will George want?

George will receive a greater net benefit if he pays less. George will prefer the Proportional Tax (% of income) because he earns half as much as Emily. This means he will only pay for 33% and Emily pays 66%.

Note: it is possible that George will not know how much Emily earns and instead vote for a Head Tax (equal 50/50 split of costs).

iii) Which taxes will result in the construction of the driveway if it costs \$2000?

Head Tax: Emily pays \$1000. George pays \$1000. As both are willing to pay \$1000, a Head Tax will result in the construction of the driveway.

Proportional Tax (% of use): Emily pays \$666.67 (33% of \$2000). George pays \$1333.33 (66% of \$2000). As George is only willing to pay up to \$1000, the driveway will not be built.

Proportional Tax (% of income): Emily pays \$1333.33 (66% of \$2000). George pays \$666.67 (33% of \$2000). As both are willing to pay these amounts a Proportional Tax (% of income) will result in the construction of the driveway.

Question 3

Consider the following to develop your understanding of how public goods can be considered in your life.

- i) Reflecting on the payment for shared goods (such as in Q2 and the driveway), what difficulties may arise regarding an individual's ability and desire to provide the good?
- ii) Now, if this was not one shared driveway between neighbours but instead new roads for the whole town, what difficulties will arise?
- iii) What ideas do you have to overcome these difficulties to achieve the socially optimal construction of shared goods (in this case, new roads)?



Thank You

Next week: Economics of
Information

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Tutor
evaluationssssssssssssssssssssss
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